

YAPT



Yet Another Plugin Tool

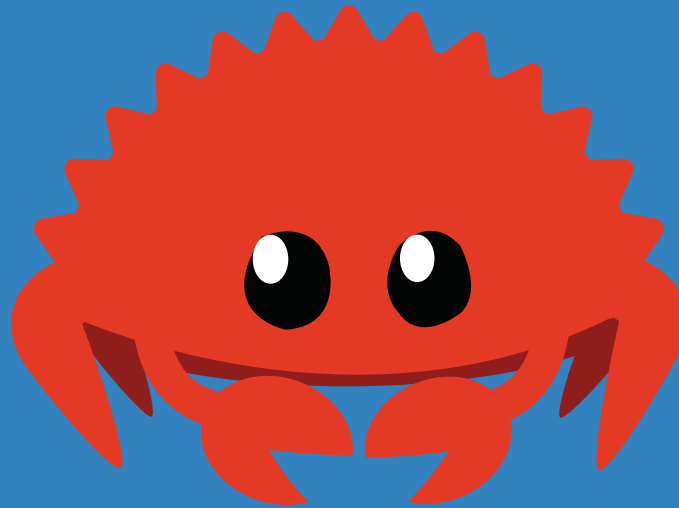
Server, manager and publisher

- qgis-plugins-server - yapt-server
- <https://github.com/3liz/yapt-manager>
- <https://github.com/3liz/yapt-package>



Disclaimer - Spoiler

To each his own!



We love CRAB!

Historic situation

Our needs

- Internal plugins repository for public and in dev plugins
- Clients plugins repository for in dev plugins
- Deploying QGIS Server plugins on our Lizmap hosting infrastructure

Do not overload `plugins.qgis.org`

Implementation

- File server
- XML files listing available plugins
- qgis-plugin-manager

qgis-plugin-manager

CLI tool to manage QGIS plugins

```
$ qgis-plugin-manager --help
usage: qgis-plugin-manager [-h] [-v] {version,init,list,install,remove,upgrade,remotes,update,cache,versions,search,check} ...

options:
  -h, --help            show this help message and exit
  -v, --verbose          Activate verbose (debug) mode (default: False)

commands:
  qgis-plugin-manager command

{version,init,list,install,remove,upgrade,remotes,update,cache,versions,search,check}
  version              Show version informations and exit
  init                 Create the `sources.list` with plugins.qgis.org as remote
  list                 List all plugins in the directory
  install              Install a plugin
  remove               Remove a plugin by its name
  upgrade              Upgrade all plugins installed
  remotes              List all remote server
  update              Update all index files
  cache                Look for available plugin in the cache - Deprecated
  versions             Look for available plugin latest versions
  search               Search for plugins
  check                Check compatibility of installed plugins with QGIS version
```

Evolution of our needs

Multiple In-dev plugins versions: -pre

For exemple:

- master with QGIS4 support
- stable Without QGIS4 support

Or:

- master with code refactoring
- stable with bugfixings only

Pushed to plugins repository for manual or automatic tests

Install specific version of a plugin even if it is not the last one

QGIS plugins server - The YAPT Server

QGIS Plugin Repository Server: It maintains a catalog of packages populated by archive uploads via HTTP.

```
curl -X POST -F "file=@monplugin.1.0.0.zip" http://localhost:8070/plugins/  
curl "http://localhost:8070/plugins.xml?qgis=3.40"
```



QGIS plugins server - The YAPT Server

Features

- **Automatic extraction of metadata** (metadata.txt) from the uploaded archive;
- Catalog updates without the **risk of concurrent access**;
- **Plugin filtering** by QGIS version, tag, and status (experimental, deprecated, server plugin);
- Catalog available in **XML (QGIS Desktop) and JSON**;
- HTTP cache management (ETag / Last-Modified);
- Optional TLS and client certificate authentication (mTLS).

It is not

- A web site like qgis.plugin.org
- A security validator (no virus scan, no code analysis)

QGIS plugins server - The YAPT Server

API HTTP - GET

```
# XML Catalog for QGIS 3.40, including experimental versions
curl "http://localhost:8070/plugins.xml?qgis=3.40&pre=true"
# Only server plugins, in JSON
curl "http://localhost:8070/plugins.json?qgis=3.40&server=true"
# All versions of a plugin
curl "http://localhost:8070/plugins/lizmap-server/plugins.json?qgis=3.40&all=true&pre=true"
```

API HTTP - POST

```
curl -i -X POST \
  -H "X-Upload-Agent: gitlab-ci" \
  -F "file=@lizmap-server.2.16.0.zip" \
  -F "checksum=@lizmap-server.2.16.0.zip.sha256" \
  "https://plugins.example.org/plugins/"
```

API HTTP - DELETE

```
curl -X DELETE "https://plugins.example.org/plugins/lizmap-server/?version=%3D2.16.0"
{"removed":[{"lizmap-server","2.16.0"}]}
```

QGIS plugins server - The YAPT Server

API HTTP - GET -Routes

Methods	Route	Description
GET , HEAD	/plugins.xml	Catalog XML (QGIS Desktop)
GET , HEAD	/plugins.json	Catalog JSON
GET , HEAD	/plugins/	Catalog JSON (alias)
GET , HEAD	/plugins/<slug>/plugins.xml	Catalog XML for only one plugin
GET , HEAD	/plugins/<slug>/plugins.json	Catalog JSON for only opne plugin
GET , HEAD	/plugins/<slug>/	Catalog JSON for only one plugin

QGIS plugins server - The YAPT Server

API HTTP - GET - Request parameters:

Parameter	Value	Default	Description
qgis	version	—	Required. Target QGIS version (3.40 , 3.40.2 , ...)
pre	true / false	false	Includes experimentals versions
deprecated	true / false	false	Includes deprecated plugins
server	true / false	false	Only server plugins
all	true / false	false	Returns all versions, JSON only
tags	texte	—	Approximate search based on the plugin's name and tags

- Boolean values must be specified explicitly: `?pre=true` . `?pre` or `?pre=1` returns `400` .
- Without `all=true` , the catalog contains no more than two versions per plugin, like **qgis.plugins.org**.

QGIS plugins server - The YAPT Server

API HTTP - POST

POST /plugins/[?pre=true|false]

multipart/form-data body:

Field	Required	Description
file	yes	The plugin's .zip archive
sig	no	Archive signature
checksum	no	Checksum file
checksum_sig	no	Checksum signature

- Optional header X-Upload-Agent : stored in the field uploadedBy
- Parameter pre overrides the experimental field of metadata.txt

QGIS plugins server - The YAPT Server

API HTTP - DELETE

```
DELETE /plugins/<slug>/?version=<requirement>
```

- `version` is a constraint in [SemVer](#) format (`=0.1.0`, `<1.0.0`, `>=1.0`, `<2.0`, ...).
- **⚠ Warning:** `version=0.1` is interpreted as `^0.1` and therefore deletes all `0.1.x` versions; use `=0.1.0` for an exact match.
- The corresponding archives are deleted from disk and the catalog is rewritten. The response lists the versions that were actually deleted.

QGIS plugins server - The YAPT Server

Not Ready Yet

Tested on ~20 plugins only
No database providers to store plugins metadata



YAPT - Now that we're here, why not keep going?



YAPT-pkg



<https://github.com/3liz/yapt-package>

YAPT-pkg - The YAPT packager

```
# Basic package creation
yapt-pkg package

# Create archive with prerelease version
yapt-pkg package --pre

# Create archive in specific directory
yapt-pkg package -o ./dist

# Generate package XML for QGIS
yapt-pkg package --xml "https://example.com/downloads/"

# Publish to QGIS plugin repository
yapt-pkg package --publish --osgeo-username "user" --osgeo-password "pass"

# Publish with environment variables
OSGEO_USERNAME=user OSGEO_PASSWORD=pass yapt-pkg package --publish

# Dry run (test connection)
yapt-pkg package --publish --dry-run
```

YAPT-pkg - The YAPT packager

```
# Get changelog for current version (from pyproject.toml)  
yapt-pkg changelog
```

```
# Get changelog for specific version  
yapt-pkg changelog --version "1.0.0"
```

YAPT-manager



<https://github.com/3liz/yapt-manager>

YAPT-manager - QGIS Plugin Manager

```
yapt-mngr --help
A QGIS plugin manager
```

```
Usage: yapt-mngr [OPTIONS] <COMMAND>
```

Commands:

```
source  Manage sources
list    List installed plugins
find    Find plugins
install Install plugin(s)
upgrade Upgrade installed plugins with remote sources
search  Search for plugins
remove  Remove installed plugins
help    Print this message or the help of the given subcommand(s)
```

Global options:

```
-C, --config <CONFIG>      The configuration directory path (default to current dir) [env: YAPT_CONF_DIR=]
--cache-dir <CACHE_DIR>    The cache directory (default to config dir) [env: YAPT_CACHE_DIR=]
--no-sync                  Do no synchronize sources [env: YAPT_NO_SYNC=]
--qgis-version <QGIS_VERSION> The QGIS version [env: QGIS_VERSION=]
-d, --install-dir <INSTALL_DIR> Plugin installation directory [env: QGIS_PLUGINPATH=/var/qjazz/plugins]
--no-progress              Hide all progress outputs [env: YAPT_NO_PROGRESS=]
-v, --verbose...           Increase log verbosity
-h, --help                 Display concise help
-V, --version               Display the program version
```

YAPT-manager - QGIS Plugin Manager

```
yapt-mngr source --help
Manage sources
```

```
Usage: yapt-mngr source [OPTIONS] <COMMAND>
```

```
Commands:
```

```
  add      Add remote source
  remove   Remove remote source
  rename   Rename source
  list     List sources
  update   Fetch sources [alias: sync]
  check    Check for update
  help     Print this message or the help of the given subcommand(s)
```

YAPT-manager - QGIS Plugin Manager

```
yapt-mngr list --help
List installed plugins
```

```
Usage: yapt-mngr list [OPTIONS]
```

```
Options:
  -o, --outdated      List outdated plugins
  --source <SOURCE>  Use only the specified source when searching for latest version
  --pre               Include pre-release, development and experimental versions [env: QGIS_PLUGIN_INCLUDE_PRERELEASE=]
```

```
yapt-mngr list
3liz.org.... ✓ Up to date
```

NAME	VERSION	QGIS*	SOURCE	LATEST	FOLDER
atlasprint	3.4.4	3.28	3liz.org	3.4.4	/srv/qgis-server-plugins/atlasprint
Lizmap server	2.15.4	3.34	3liz.org	2.15.5	/srv/qgis-server-plugins/lizmap_server
wfsOutputExtension	1.8.3	3.28	3liz.org	1.8.3	/srv/qgis-server-plugins/wfsOutputExtension

(*) Minimum QGIS versions supported

```
yapt-mngr list --outdated
3liz.org.... ✓ Up to date
```

NAME	VERSION	QGIS*	SOURCE	LATEST	FOLDER
Lizmap server	2.15.4	3.34	3liz.org	2.15.5	/srv/qgis-server-plugins/lizmap_server

(*) Minimum QGIS versions supported

YAPT-manager - QGIS Plugin Manager

```
yapt-mngr find --help
Find plugins
```

```
Usage: yapt-mngr find [OPTIONS] <NAME>...
```

```
Arguments:
  <NAME>...  List of plugins with optional version specifiers
```

```
Options:
  --pre            Include pre-release, development and experimental versions [env: QGIS_PLUGIN_INCLUDE_PRERELEASE=]
  --deprecated     Include deprecated versions
  --source <SOURCE> Use only the specified source
```

Find lizmap plugin with version 5.x.x from 3liz.org QGIS plugins server.

```
yapt-mngr find lizmap=5
3liz.org.... ✓ Up to date
lizmap:
NAME    VERSION QGIS SOURCE
Lizmap  5.0.3   3.34 3liz.org
Lizmap  5.0.2   3.34 3liz.org
Lizmap  5.0.1   3.34 3liz.org
Lizmap  5.0.0   3.34 3liz.org
```

YAPT-manager - QGIS Plugin Manager

```
yapt-mngr search --help
Search for plugins
```

```
Usage: yapt-mngr search [OPTIONS] <NAME>
```

```
Arguments:
  <NAME>
```

```
Options:
  --by-name           Only search by plugin name
  --server            Consider only server plugins
  --all               Return all versions of plugins
  --pre               Include pre-release, development and experimental versions [env: QGIS_PLUGIN_INCLUDE_PRERELEASE=]
  --deprecated        Include deprecated versions
  --source <SOURCE>  Use only the specified source
```

```
yapt-mngr search lizmap,server
3liz.org.... ✓ Up to date
 S Server  X  Experimental  T  Trusted  D  Deprecated
NAME          VERSION QGIS* STATUS SOURCE
Lizmap         5.0.3   3.34  ----  qgis-plugins.3liz.org
Lizmap server  2.15.5  3.34  S---  qgis-plugins.3liz.org
wfsOutputExtension 1.8.3   3.28  S---  qgis-plugins.3liz.org
(*) Minimum QGIS versions supported
```

YAPT-manager - QGIS Plugin Manager

```
yapt-mngr install --help
Install plugin(s)

Usage: yapt-mngr install [OPTIONS] <NAME>...

Arguments:
  <NAME>...  Plugins to install

Options:
  --pre            Include pre-release, development and experimental versions [env: QGIS_PLUGIN_INCLUDE_PRERELEASE=]
  --deprecated     Include deprecated versions
  --source <SOURCE> Use only the specified source
  --reinstall      Force (re)installing
  --fix-permissions Set files permissions to 0644
  -U, --upgrade    Upgrade plugin to latest version, if '--pre' is specified, the update will update to the latest experimental version if any
  --dry-run        Only show what would be installed
```

YAPT-manager - QGIS Plugin Manager

```
yapt-mngr upgrade --help
Upgrade installed plugins with remote sources
```

```
Usage: yapt-mngr upgrade [OPTIONS]
```

Options:

--pre	Include pre-release, development and experimental versions [env: QGIS_PLUGIN_INCLUDE_PRERELEASE=]
--deprecated	Include deprecated versions
--source <SOURCE>	Use only the specified source
--reinstall	Force (re)installing
--fix-permissions	Set files permissions to 0644
--dry-run	Only show what would be installed

YAPT-manager - QGIS Plugin Manager

```
yapt-mngr remove --help  
Remove installed plugins
```

```
Usage: yapt-mngr remove [OPTIONS] <NAME>...
```

Arguments:

```
<NAME>...  Plugins to remove
```

Conclusion

- A set of tools written in **Rust** to manage QGIS plugins
- Nothing new - **Yet Another Plugin Tool**



Thank you



Questions ?